Solutions - Practice Problem Set 5

ECON 101 – Macroeconomics

Fall 2025

Part A – Multiple Choice Solutions

1. Correct Answer: (C) A lower price level increases consumption, investment, and net exports — causing a movement *down* along the AD curve.

2. Correct Answer: (C) A depreciation makes Canadian exports cheaper and imports more expensive, increasing net exports and shifting AD right.

3. Correct Answer: (B) A higher price level increases firms' profits only in the short run, causing a *movement along* SRAS, not a shift.

4. Correct Answer: (D) Money multiplier:

$$m = \frac{1}{rr} = \frac{1}{0.10} = 10$$

A new \$10,000 deposit leads to a maximum increase of:

$$\Delta M = 10{,}000 \times 9 = 90{,}000$$

5. Correct Answer: (A) An open-market purchase increases reserves, increases the money supply, and lowers the overnight interest rate.

Part B – Long Analytical Problem Solutions

6. Monetary Policy and AD–AS Interaction

(a) Short-run effects in the AD–AS model

A drop in consumer confidence reduces autonomous consumption. This shifts the AD curve leftward:

- Real GDP falls below potential GDP ($Y < Y^*$). - The price level falls due to reduced demand. - A recessionary gap opens.

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(b) Change in GDP with MPC = 0.8

Initial shock:

$$\Delta A = -100$$

Multiplier:

$$k = \frac{1}{1 - MPC} = \frac{1}{1 - 0.8} = 5$$

Total effect on GDP:

$$\Delta Y = k \cdot \Delta A = 5 \times (-100) = \boxed{-500}$$

GDP falls by $500 billion in total.

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(c) Bank of Canada policy

To close the recessionary gap, the Bank of Canada conducts an **open-market purchase of government securities**. This:

- Increases bank reserves - Expands the money supply - Lowers interest rates - Stimulates consumption and investment - Shifts AD rightward back to potential GDP

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(d) Required OMO size with money multiplier = 5

Let ΔR = change in reserves from the Bank of Canada's purchase.

Money multiplier:

$$m = 5$$

We need to offset the fall in spending:

$$\Delta A = +100$$

To prevent the multiplied effect of -500, we need:

$$\Delta Y = +500$$

Since $\Delta M = m \cdot \Delta R$, and an increase in money stimulates AD one-for-one in this simplified setup:

$$\Delta M = 500$$

Thus:

$$\Delta R = \frac{500}{5} = \boxed{100}$$

The Bank of Canada needs a **$100 billion open-market purchase** of bonds.

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End of Instructor Solution Key